

CS & AI • EXPLAINED

How does your phone know your **location**?

Your phone has no idea where it is — until it listens to clocks in space.

THE HOOK

You open Maps and a little **blue dot** knows **exactly** where you are.

But your phone has no built-in sense of place. So... how?

It works it out by listening to satellites — clocks floating in space. 🛰️



TODAY'S QUESTION

With no GPS chip telling it,
how does your phone **figure**
out where it is on **Earth**?

Satellites just **broadcast**, all the time.



Satellite

))) signal)))



Everyone

About **30** satellites orbit Earth, shouting to the whole planet.

- Each one constantly broadcasts **two things**.
- Its own **position** in space.
- The exact **time** from an ultra-precise **atomic clock**.

How **long** it took → how **far** it is.

- Your phone catches signals from **several** satellites.
- Each signal travels at the **speed of light**.
- Measure exactly how long it took to arrive...
- ...and the phone knows how **far** that satellite is.

THE TRICK

time delay
→ **distance**

One spot fits **all of them**.

- One distance puts you *somewhere* on a giant **sphere**.
- Add a second, then a third...
- ...and only **ONE point** on Earth fits all the distances at once. That's you.

LIKE THREE FRIENDS

Each says "you're 5 km from me." Only **one spot** satisfies all three.

This is called **trilateration**.

It only **listens**. Never transmits.



Satellites

))) one way)))



Phone

Your phone **receives** only — it never sends anything *up*.

- It needs about **4** satellites.
- **3** to pin the position...
- ...**1** more to correct the phone's own clock.
- **More satellites** = more accuracy.

WHY IT MATTERS

This blue dot runs your day.

- Maps, **Uber / Ola**, food delivery, fitness trackers — all ride on GPS.
- Indoors, in tunnels, or between **tall buildings**? Signals get blocked — so GPS gets weak.
- Constantly listening to faint space signals is also why GPS can **drain your battery**.

COMMON MYTH — "YEH GALAT SAMAJHTE HAIN"

X MYTH

"Satellites **track and watch me** — my phone **talks** to the satellites."

✓ REALITY

Satellites just **broadcast** their time and position to the whole world. Your phone silently **listens** and calculates its own location. The satellites don't even know you exist.

RECAP — 2 THINGS TO REMEMBER

- 1 GPS satellites just **broadcast** their time and position to everyone.
- 2 Your phone **times** those signals to get distances, then **trilaterates** to pin your exact spot.

GPS = listening to clocks in space. 

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Next question: What is a programming language, and why so many? 